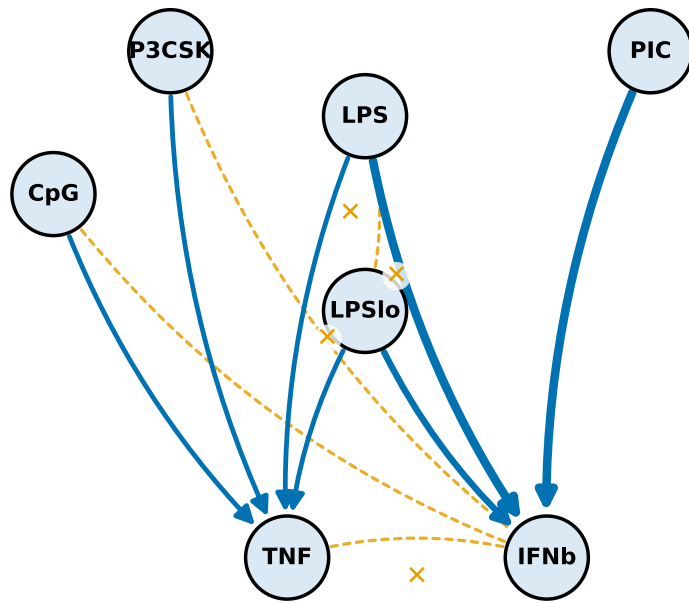
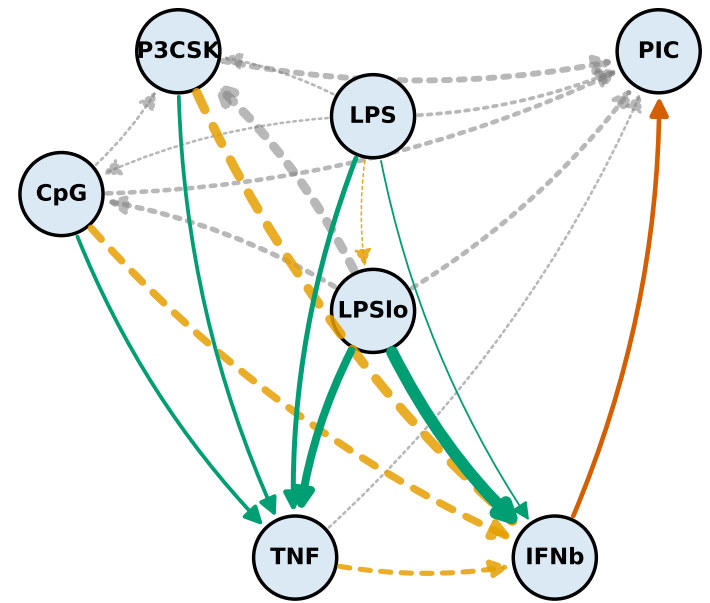


Sheu 2024 BMDM: real-data cascade-direction validation (bars explicitly scaled by $|\text{cross_asym}|$)

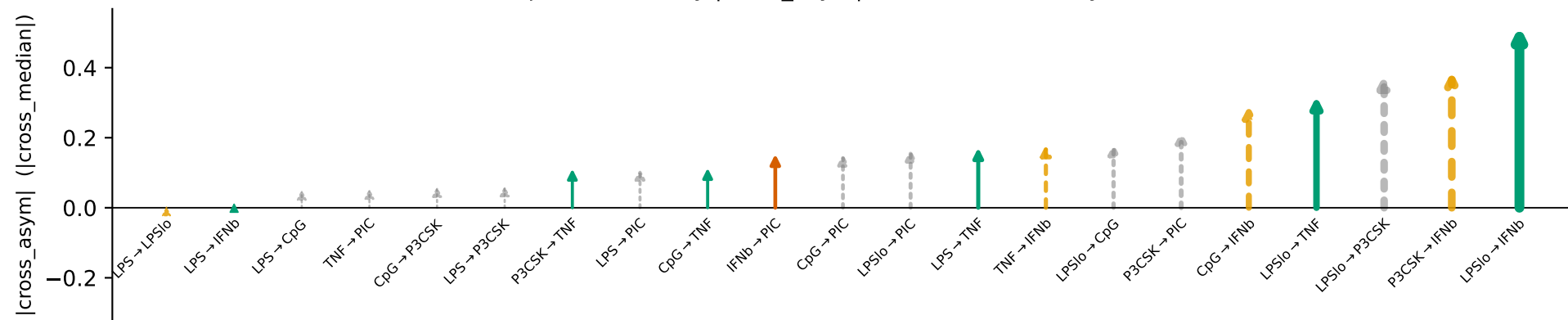
(a) Pre-registered ground truth
(Sheu TLR cascade wiring, §21/§26)



(b) What cross_asym found (Sheu, 5hr)
benchmark direction accuracy: $6/7 = 86\%$



(c) All 21 pairs, sorted by $|\text{cross_asym}|$ (direction carried by the tick label)



- ground truth cascade edge
- ground truth: no cascade (negative control)
- found: correct direction (benchmark pair)
- found: wrong direction (benchmark pair)
- found: no literature prior (uncalibrated)